

- 2** The variables x and y are related by the differential equation

$$6\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + x = t^2 + 10t + 13.$$

- (a) Find the general solution for x in terms of t . [6]

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- (b) State an approximate solution for large positive values of t . [1]

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